IEOR 4404: Simulation

Lecture 11: Simulating Poisson Processes

Counting Process

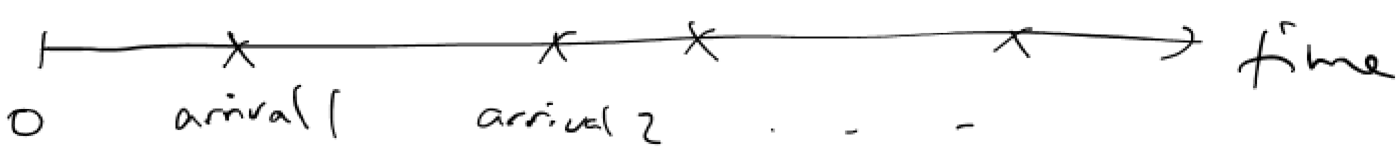
Model the counts of events

E.g., customer arrivals, insurance claims..

A counting process satisfies:

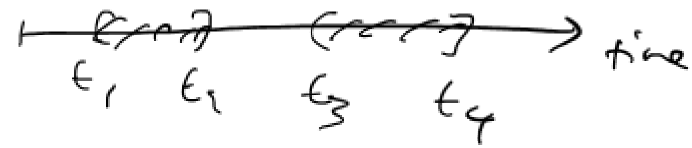
- $N(t) \geq 0$
- $N(t)$ is integer-valued
- $N(t)$ is non-decreasing
- $N(t) - N(s)$ counts the number of events between $(s, t]$

$N(t) =$ # arrivals before time t



Poisson Process

A counting process with the following properties:



- $N(0) = 0$
- $N(t)$ has independent increments
- $P(N(t+h) - N(t) = 1) = \lambda h + o(h)$
- $P(N(t+h) - N(t) \geq 2) = o(h)$

$N(t_2) - N(t_1)$ and $N(t_4) - N(t_3)$ are independent.

a function of h, say $f(h)$, such that $\frac{f(h)}{h} \to 0$ as $h \to 0$.

The parameter λ is called the rate

The Poisson process has both independent and stationary increments

Intuitively, $o(h)$ is negligible compared to λh when h is small.

Poisson Process

For a Poisson process with rate λ, its increment over a length t is distributed according to $Poi(\lambda t)$

This is from the Poisson approximation to binomial, or the "law of small numbers"

$$N(t) \sim Poi(\lambda t)$$

law of small numbers

$Bin(n, p)$, if n is big, p is small, so that $np = \lambda$, then

$$Bin(n,p) \approx Poi(\lambda)$$

$$P(N(t+h) - N(t) = 1) \approx \lambda h$$

$$P(N(t+h) - N(t) \geq 2) \approx 0$$

$$P(N(t+h) - N(t) = 0) \approx 1 - \lambda h$$

$$\Rightarrow \quad N(t+h) - N(t) \approx \mathrm{Ber}(\lambda h)$$

$$N(t) \approx \mathrm{Bin}\left(\frac{t}{h}, \lambda h\right) \approx \mathrm{Poi}(\lambda t)$$

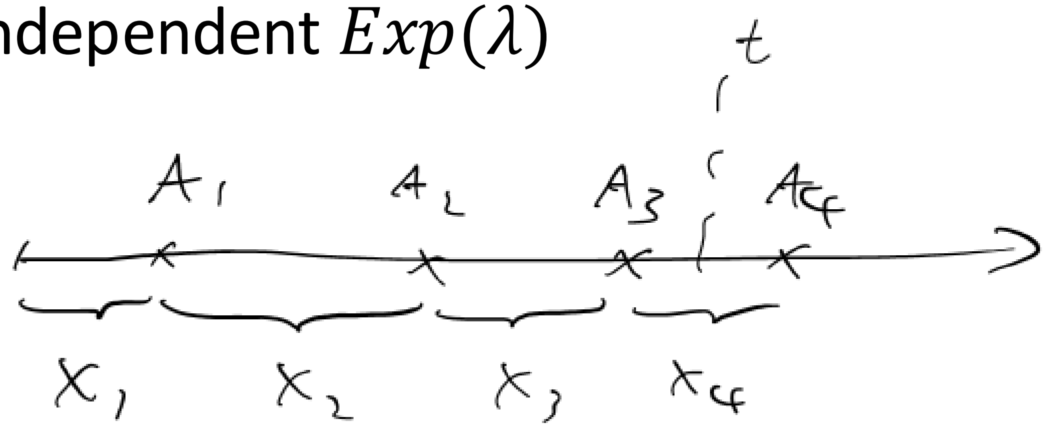
Interarrival Times

Let $A_i, i = 1,2, \ldots$ be the arrival times of events. Then

$$N(t) = \max\{n: A_n \leq t\}$$

Both $N(t), t \geq 0$ and $A_i, i = 1,2, \ldots$ can keep track of a counting process.

For a Poisson process, the interarrival times $X_n = A_n - A_{n-1}$ follow independent $Exp(\lambda)$



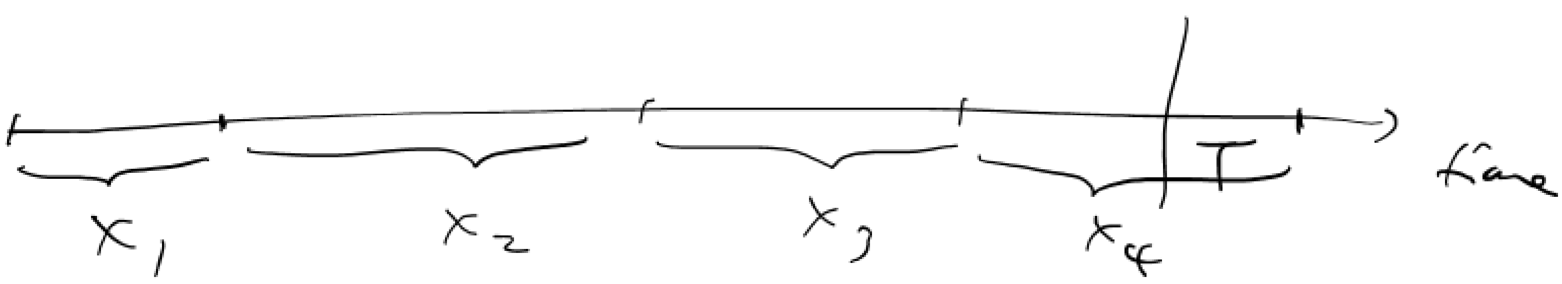
Simulating Poisson Process

To simulate a Poisson process, we can generate the interarrival times via exponential variables

To generate $N(t), t \leq T$:

- Generate $X_i \sim Exp(\lambda), i = 1, \dots, n+1$ until we see $X_1 + \cdots + X_n \leq T, X_1 + \cdots + X_n + X_{n+1} > T$
- Then $A_i = X_1 + \cdots + X_i, i = 1, \dots, n$

The outputs encode the arrival times of all events up to time T



Simulating Poisson Process

Pseudo code:

arrival count

Set $t = 0, I = 0$ time of the latest arrival

While $t \leq T$,

- Generate $U \sim Unif(0,1)$
- $t = t - \frac{1}{\lambda} \log U$

ITM for $Exp(\lambda)$

- $I = I + 1$, $A(I) = t$

arrival time of the latest arrival

Output $\underbrace{I - 1}_{N(T)}$ and $\underbrace{A(i), i = 1, \dots, I - 1}_{\text{arrival times before } T.}$

Conditional Representation

The conditional distribution of $A_1, \dots, A_n$ given $N(t) = n$ is the same as the order statistic of n independent $Unif(0, t)$

Order statistic:

Generate n independent $Unif(0, t)$, and sort the values in increasing order to obtain

$$\left(U_{(1)}, \dots, U_{(n)}\right)$$

This is the order statistic

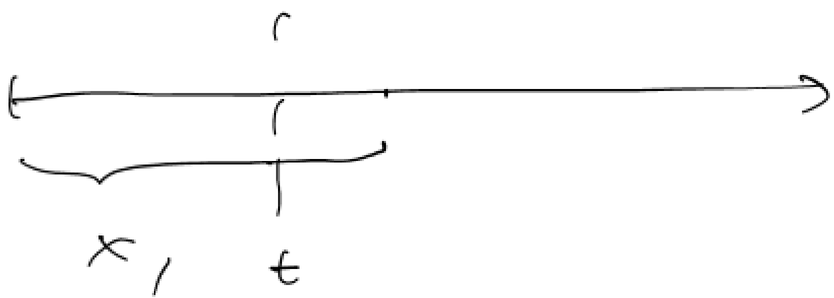
Alternative Simulation Approach

To simulate a Poisson process over $[0, T]$,

- Generate $N(T) \sim Poi(\lambda T)$
- Suppose $N(T) = n$. Generate n i.i.d. $Unif(0,1)$, say $U_1 , \ldots , U_n$
- Sort $U_1, \ldots , U_n$ in increasing order to get $U_{(1)} < \cdots < U_{(n)}$
- Output n and $A_i = TU_{(i)}, i = 1, \ldots , n$

Explanation of interarrival time $\sim \text{Exp}(\lambda)$:

$$P(X_1 > t) = P(N(t) = 0) = \frac{e^{-\lambda t}(\lambda t)^0}{0!} = e^{-\lambda t}$$



and reiterate this argument for $x_2, x_3, \ldots$

and also argue they are indept by indept increment property of Poisson process.

Explanation of conditional representation:

We want to find dist of $A_1, A_2, \dots, A_n$

given $N(t) = n$

To do so, consider the density of $A_1, \dots, A_n$

given $N(t) = n$

$$f(t_1, t_2, \dots, t_n \mid N(t) = n)$$

$$f(t_1, \dots, t_n \mid N(t) = n)$$

$$= \frac{P(A_1 = t_1, \dots, A_n = t_n, A_{n+1} > t)}{P(N(t) = n)}$$

$$= \frac{P(X_1 = t_1, X_2 = t_2 - t_1, \dots, X_n = t_n - t_{n-1}, X_{n+1} > t - t_n)}{P(N(t) = n)}$$

$$= \frac{\lambda e^{-\lambda t_1} \cdot \lambda e^{-\lambda(t_2 - t_1)} \cdots \lambda e^{\lambda(t_n - t_{n-1})} \cdot e^{-\lambda(t - t_n)}}{\frac{e^{-\lambda t}(\lambda t)^n}{n!}}$$

$$= n! \frac{1}{t^n}$$

This is the density of order statistic of

n Unif$(0, t)$